



FACTORY METHOD

```
package com.bibliotech.factorymethod;
```

```
• public class MediaPonderadaFactory extends MediaFactory {  
  
    private double[] pesos;  
  
    public MediaPonderadaFactory(double[] pesos) {  
        this.pesos = pesos;  
    }  
  
    @Override  
    public MediaMetricas criarMedia() {  
        return new MediaPonderada(pesos);  
    }  
• }
```

```
package com.bibliotech.factorymethod;
```

```
public class MediaAritmeticaFactory extends MediaFactory {  
  
    @Override  
    public MediaMetricas criarMedia() {  
        return new MediaAritmetica();  
    }  
}
```

```
package com.bibliotech.factorymethod;
```

```
public class MediaPonderada implements MediaMetricas {
```

```
    private double[] pesos;
```

```
    public MediaPonderada(double[] pesos){
```

```
        this.pesos = pesos;
```

```
    }
```

```
    @Override
```

```
    public double calcular(double[] notas) {
```

```
        double soma = 0;
```

```
        double somaPesos = 0;
```

```
        for(int i = 0; i < notas.length; i++){
```

```
            soma += notas[i] * pesos[i];
```

```
            somaPesos += pesos[i];
```

```
        }
```

```
        return soma / somaPesos;
```

```
    }
```

```
}
```

```
package com.bibliotech.factorymethod;
```

```
public class MediaAritmetica implements MediaMetricas {
```

```
    @Override
```

```
    public double calcular(double[] notas) {
```

```
        double soma = 0;
```

```
        for(double nota : notas){
```

```
            soma += nota;
```

```
        }
```

```
        return soma / notas.length;
```

```
    }
```

```
}
```

```
package com.bibliotech.factorymethod;

public interface MediaMetricas {
    double calcular(double[] notas);
}
```

```
package com.bibliotech.factorymethod;

public abstract class MediaFactory {
    public abstract MediaMetricas criarMedia();
}
```

```
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public class Main {

    public static void main(String[] args) {

        double[] notas = {8.0, 7.5, 9.0};

        /*MediaFactory factory = new MediaAritmeticaFactory();

        MediaMetricas media = factory.criarMedia();

        System.out.println("Média Aritmética: " + media.calcular(notas));*/

        double[] pesos = {2, 3, 5};

        MediaFactory factory = new MediaPonderadaFactory(pesos);

        MediaMetricas media = factory.criarMedia();

        System.out.println("Média Ponderada: " + media.calcular(notas));
    }
}
```


